

Comparison of Longley-Rice, ITU-R P.1546 and Hata-Davidson propagation models for DVB-T coverage prediction

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Abstract— This paper compares precision field-strength measurements taken by a Rohde & Schwarz FSH-3 portable spectrum analyzer with simulation results derived from the ITM coverage prediction model (Irregular Terrain Model), also known as NTIA-ITS Longley-Rice model, in conjunction with the 3-arc-second SRTM (Satellite Radar Topography Mission) geographical data, the propagation predictions of ITU-R Recommendation P.1546 and those of the empirical Hata-Davidson model using HAAT. ITU-R P.1546 and Hata-Davidson models exhibit higher errors at longer distances and therefore necessary corrections should be introduced in the models in order to increase propagation prediction accuracy. Especially, measurements results show that ITU-R P.1546, on average, underestimates the field strength at distances longer than 50 km. The Longley-Rice model using the terrain digital elevations is more accurate, as expected, and its results are closer to the measurement data.

Keywords—DVB-T, HAAT, Hata-Davidson model, ITM, ITU-R.1546, Longley-Rice, NTIA-ITS, , SRTM.

I. INTRODUCTION

The huge increase in designing broadcast systems for digital TV and cellular systems necessitates the development of accurate point-to-area prediction tools. In electromagnetic wave propagation theory there is a great number of coverage prediction models for DVB-T coverage and, more specifically, there are three main types of propagation models: empirical models that are based on measurement data, they are simple and use statistical data but they are not very accurate, semi-deterministic models that are based on empirical data and deterministic aspects, and deterministic models that are based on theory and require a lot of geometrical information about the site and the terrain, need a very significant computational effort, but they are more accurate.

The aim of this research is to provide coverage prediction results for DVB-T services, in the region of Thessaloniki – Greece, and compare the simulation results produced by the propagation prediction software with field measurements taken by our measurement equipment. It is well known that

the use of prediction models becomes a necessity, because making on-site measurements costs in time and money.

The Radio Mobile program [1-2], (Radio Propagation and Virtual Mapping Freeware) is based on the Longley-Rice Model ITM [3] and uses the 3-arc-second Satellite Radar Terrain Mission SRTM maps, [4]. The Radio communication Sector of ITU proposed the Recommendation ITU-R P.1546, [5]. It is a method for point-to-area-prediction for terrestrial services in the frequency range 30MHz to 3GHz. This method is based on empirically derived field-strength curves and has replaced the older ITU-R 370 Recommendation. These curves are functions of frequency, antenna height, distance and time percentage. Extrapolation and interpolation methods are used to derive data from these empirical curves. The field strength (dBμV/m) is calculated for 1 kW ERP (Effective Radiated Power). The latest version is ITU-R P.1546-5, Sept. 2013.

The empirical Hata model, [6], on the other hand, uses mathematical equations that involve antenna heights of transmitter and receiver, frequency 150MHz to 1500MHz, distance from the base station ranges from 1Km to 20Km in urban, suburban and open area environments. It is an extension of Okumura's method and it is based on Okumura's field test results for the city of Tokyo, Japan, [7-9].

The Telecommunications Industry Association (TIA) recommended a modification to the Hata model in order to cover a broader range of input parameters and distances. The model is known as Hata-Davidson model and provides corrections for links up to 300km distance and transmitters at an altitude up to 2500m. It was published in TSB-88A, [10].

The formulas of the Hata-Davidson model are:

$$PL_{HD} = PL_{Hata} + A(h_1, d_{km}) - S_1(d_{km}) - S_2(h_1, d_{km}) - S_3(f_{MHz}) - S_4(f_{MHz}, d_{km}).$$

where: $PL_{Hata} = 69.55 + 26.16 \log_{10} f_{MHz} - 13.82 \log_{10} h_1 - a(h_2) + (44.9 - 6.55 \log_{10} h_1) \log_{10} d_{km}$
 $a(h_2)$, is the correction factor for the mobile antenna height:

$$a(h_2) = \begin{cases} (1.1 \log_{10} f_{MHz} - 0.7) * h_2 - (1.56 \log_{10} f_{MHz} - 0.8) & \text{for medium small city, quasi open, open area} \\ 8.29 (\log_{10} (1.54 * h_2))^2 - 1.1 & \text{for large city and } f_c \leq 300 \text{ MHz} \\ 3.2 (\log_{10} (11.75 * h_2))^2 - 4.97 & \text{for large city and } f_c > 300 \text{ MHz} \end{cases}$$

where,

- f_{MHz} : is the carrier frequency (30-1500MHz.)
- h_1 : is the base station antenna height HAAT (20-2500m).
- h_2 : is the mobile station antenna height (1-10m).
- d_{km} : is the transmission distance (1-300km).
- A and S_1 : are factors that extend the distance to 300Km
- S_2 : is a correction factor for the height h_1 of the base station antenna extending the value of h_1 to 2500m,
- S_3 and S_4 : are correction factors that extend the frequency to 1500MHz.

Definitions of the terms are given below.

Distance d_{km} in Km	$A(h_1, d_{km})$	$S_1(d_{km})$
$d_{km} < 20$	0	0
$20 \leq d_{km} < 64.38$	$0.62137 (d_{km} - 20)[0.5 + 0.15 \log_{10}(h_1/121.92)]$	0
$64.38 \leq d_{km} < 300$	$0.62137 (d_{km} - 20)[0.5 + 0.15 \log_{10}(h_1/121.92)]$	$0.174(d_{km} - 64.38)$

$$S_2(h_1, d_{km}) = 0.00784 |\log_{10}(9.98/d_{km})| (h_1 - 300) \text{ for } h_1 > 300m$$

$$S_3(f_{MHz}) = f_{MHz}/250 \log_{10}(1500/f_{MHz})$$

$$S_4(f_{MHz}, d_{km}) = [0.112 \log_{10}(1500/f_{MHz})] (d_{km} - 64.38) \text{ for } d_{km} > 64.38 \text{ Km.}$$

Field strength is calculated from the formula below:

$$E(\text{dB}\mu\text{V/m}) = \text{ERP (dBW)} - \text{PL}_{\text{HD}} (\text{dB}) + 20 \log_{10} f_{\text{MHz}} + 109.35$$

Free Space Loss is calculated from the well known formula:

$$\text{PL}_{\text{FREE SPACE LOSS}} = 32.44 + 20 \log_{10}(f_{\text{MHz}}) + 20 \log_{10}(d_{\text{km}}).$$

Field-strength is not allowed to assume values higher than the free-space values.

II. MEASUREMENTS AND COMPARISONS

In order to measure the signal strength of DVB-T transmissions, a measurement campaign was carried out around the city of Thessaloniki, [11-12], located in the north of Greece. A point-to-point analysis for Greek public TV ERT, Channel UHF 23 ($f = 490\text{MHz}$ & $P_t = 1600\text{W} = 32\text{dBW}$ and $\text{ERP} = 42\text{dBW}$) is made for the measurement points shown in Table I.

One of main correction factors that Davidson added to Hata's formulas is the HAAT factor (antenna Height Above Average Terrain). In the Hata model formulas are limited in their applicability to:

- Range from Base, 1-20 Km.
- Frequency Range, 150-1500 MHz.
- Base HAAT, 20-200 meters.

In the Hata-Davidson model these limits are extended to:

- Range from Base, 1-300 Km.
- Frequency Range, 30-1500 MHz.
- Base HAAT, 20-2500 meters.

Table I. Measurements points and HAAT.

No.	Measurements Points	d(km)	LAT	LONG	DEGS	HAAT (m)
1	PROFITIS ELIAS	7	40.640411	23.039927	313	487.2
2	THESSALONIKI	12.3	40.615822	22.955735	279	784.8
3	LAKE VOLVI	14.3	40.707102	23.188914	31	788.8
4	PEREA	16.7	40.513489	22.937471	236	815.5
5	METHONI	47	40.469402	22.574711	252	799.7
6	KORINOS	52	40.307130	22.618620	232	810.7
7	BORDER EVZONI	68.8	41.081410	22.588160	321	479.5
8	POLIKASTRO	69	41.081190	22.588360	320	475.9
9	SOUMELA	86	40.410086	22.116606	256	789.8
10	LOUTRAKI	107	40.966160	21.923630	293	629.4

There are many programs that can calculate HAAT. Radio Mobile (that uses the Longley-Rice model) is also one of them. In this paper, HAAT was calculated using NAD83/WGS84, FCC's terrain database, and the GLOBE 1 km database, [13]. We should clarify that the term "HAAT" refers to the HAAT in the direction of the radial under consideration and not to the overall site HAAT. HAAT values are shown in Table I and Fig. 1 for the 10 measurements points.



Fig. 1. Measurement points and HAAT.

Propagation curves of Recommendation ITU-R P.1546 that are used for the calculation of field strength, with the proper interpolations, are shown in Fig. 2. We have used these curves and the appropriate equations to obtain our calculated results. The propagation curves are designed for 1 kW effective radiated power (ERP) and the ERP of the transmitter under study is 16 kW, so that 12dB have to be added to the values of field strength produced by the propagation curves.

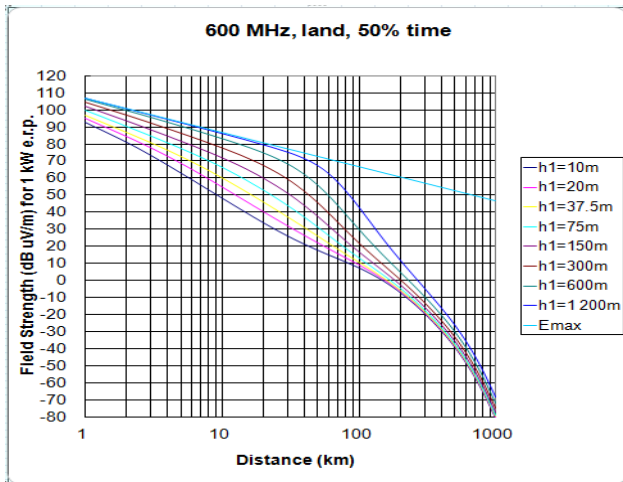


Fig. 2. Sample propagation curves of Recommendation ITU-P.1546.

In our data, produced by the ITU-R P.1546 curves, we used paths and elevation profiles from Google Earth to compute the heights H_{eff} (m) and H_b (m). Data for H_{eff} , H_a , H_b , and Path Type are shown in Table II. The time percentage is assumed to be 50%, and the receiver antenna altitude is 10m.

Table II. Values of H_{eff} , H_a , H_b , and Path Type.

No.	Measurements Points	d (km)	DEGS	H_{eff} (m)	H_a (m)	H_b (m)	Path Type
1	PROFITIS ELIAS	7	313	441.0	940	392	Land
2	THESSALONIKI	12.3	279	737.0	940	696.5	Land
3	LAKE VOLVI	14.3	31	629.5	940	618.5	Cold Sea
4	PEREA	16.7	236	722.5	-	-	Cold Sea
5	METHONI	47	252	737.0	-	-	Cold Sea
6	KORINOS	52	232	712.5	-	-	Cold Sea
7	BORDER EVZONI	68.8	321	526.5	-	-	Land
8	POLIKASTRO	69	320	526.0	-	-	Land
9	SOUVELA	86	256	740.0	-	-	Cold Sea
10	LOUTRAKI	107	293	703.5	-	-	Land

Measurements and simulations are shown in Table III. We have to mention that for our measurements the equipment used consists of a Rohde & Schwarz FSH-3, portable spectrum analyzer with tracking generator (100 kHz – 3 GHz), factory calibrated with ± 0.7 dB accuracy, two high-precision calibrated biconical antennas by Schwarzbeck, SBA 9113 (500MHz - 3GHz) and BBVU 9135 (30MHz - 1000MHz), a log-periodic precision calibrated Schwarzbeck antenna (0.25 – 6 GHz), all factory calibrated to within ± 1.0 dB accuracy, low-loss cable Huber-Suhner GX-07272-D 50 Ohm, 1.8m long, with N-type low-loss connectors.

Table III. Measurements and simulations.

No.	DVB-T Measurements Points CH23- 490MHz LAT: 40.597648 LONG: 23.0997993	LAT LONG	E(dBμV/m)			
			FSH-3	ITM Longley-Rice Rec. (0.5-2.5 m)	ITU-R P.1546 Rec. 10m	HATA-DAVIDSON With HAAT Rec. 1.5m
1	PROFITIS ELIAS (7Km/313degs)	40.640411 23.039927	101.9	95.3	96.1	97.8
2	THESSALONIKI (12,3Km/279degs)	40.615822 22.955735	94.7	98.2	93.5	95.6
3	LAKE VOLVI (14,3Km/31degs)	40.707102 23.188914	98.2	97.0	95.8	94.2
4	PEREA (16,7Km/236degs)	40.513489 22.937471	98.2	94.8	94.4	93.1
5	METHONI (47Km/252degs)	40.469402 22.574711	89.1	82.4	84.2	72.9
6	KORINOS (52Km/232degs)	40.307130 22.618620	83.5	80.4	82.8	70.5
7	BORD. EVZONI (68,8Km/321degs)	41.081410 22.588160	64.1	72.3	55.4	73.8
8	POLIKASTRO (69Km/320degs)	41.081190 22.588360	71.9	72.4	55.3	73.7
9	SOUVELA (86Km/256degs)	40.410086 22.116606	77.9	77.9	71.5	80.2
10	LOUTRAKI (107Km/293degs)	40.966160 21.923630	77.3	74.7	43.0	78.3

The Hata-Davidson computed loss is not allowed to be less than the Free-Space loss. Errors between measurements (FSH-3) and simulations from the Longley-Rice ITM model (Radio Mobile), from the ITU-R P.1546-5 model and from the Hata-Davidson model with average error, average absolute error, and standard deviation are shown in Table IV and Fig. 4.

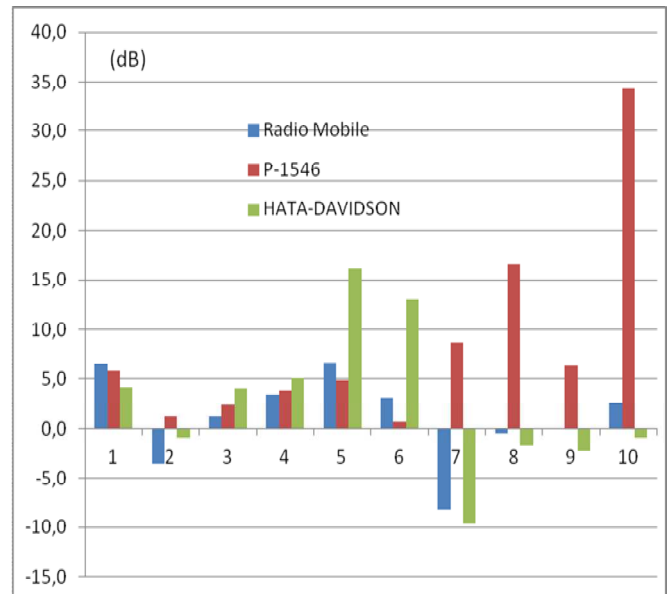


Fig. 4. Errors between measurements and simulations.

Table IV. Errors, mean error, mean absolute error, and standard deviation.

No.	DVB-T Measurements Points CH23 - 490MHz LAT : 40.597648 LONG : 23.0997993	Error (dB) FSH-3 & Longley- Rice	Error (dB) FSH-3 & P.1546	Error (dB) FSH-3 & Hata - Davidson
1	PROFITIS ELIAS (7Km/313degs)	6.6	5.8	4.1
2	THESSALONIKI (12,3Km/279degs)	-3.5	1.2	-0.9
3	LAKE VOLVI (14,3Km/31degs)	1.2	2.4	4.0
4	PEREA (16,7Km/236degs)	3.4	3.8	5.1
5	METHONI (47Km/252degs)	6.7	4.9	16.2
6	KORINOS (52Km/232degs)	3.1	0.7	13.0
7	BORDER EVZONI (68,8Km/321degs)	-8.2	8.7	-9.7
8	POLIKASTRO (69Km/320degs)	-0.5	16.6	-1.8
9	SOUVELA (86Km/256degs)	0.0	6.4	-2.3
10	LOUTRAKI (107Km/293degs)	2.6	34.3	-0.9
	Mean	1.1	8.5	2.7
	Mean absolute	3.6	8.5	5.8
	Standard Deviation	4.5	10.2	7.6

As expected, the Longley-Rice (ITM) model, using accurate terrain elevation data, is the most accurate but also the most computationally intensive, [11-12]. The ITU-R 1546-5 model while it is very accurate for shorter distances below 50 km, it becomes increasingly inaccurate for longer distances, severely underestimating the received signal strength, [14-17]. Finally, the Hata-Davidson model using HAAT, while being quite simple to implement and based on analytical approximate equations, is accurate enough, and on average more accurate than the ITU-R 1546 model. It could also be greatly improved by using different approximations for land paths and sea paths as in the ITU model.

III. CONCLUSIONS

The Longley-Rice (ITM) model using worldwide SRTM 3-arc-second data for terrain elevation gives satisfactory results in comparison to the measurements, but is more computationally intensive than simple empirical models. The propagation curves in Recommendation ITU-R P.1546 give significant errors, mainly at distances longer than 50 km and need corrections, and the Hata-Davidson model with HAAT, despite being a very simple model, gives in our case reasonable results, and at longer distances better than the ITU-R P.1546 model.

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